

Suraj Kumar Mahto

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Experience

Software Development Services - Contractor

Remote

Tether (tether.io 🌐)

May 2024 – March 2025

- Working on a push notification architecture using [UnifiedPush](#) 🌐 for Linux smartphones that supported edge LLMs.
- Continued working on the same project under [Tether](#) 🌐 (stablecoin cryptocurrency company)

Software Engineer

Remote

PearTree

April 2023 – April 2024

- Contributing upstream to the [Phosh](#) 🌐 (phone shell) ecosystem made by Purism SPC and maintained by GNOME. Working on a mobile OS, for a peer-to-peer app ecosystem based on [holepunch.to](#) 🌐 stack

Graduate Engineer Trainee

Mumbai, India

National Payments Corporation of India(NPCI)

Sept 2022 - Feb 2023

- My work was related to Site Reliability and Chaos Engineering(SRE). Project- Automated Reliability Tool.
- Worked on with deployment architectures of NETC(FastTag), RuPay, UPI that impacted systems handling **10 million+ transactions** per day

Google Summer of Code student

Remote

KDE Community

June 2021 - Aug 2021

- [Project](#) 🌐 Add feature in [KMyMoney](#) 🌐: Integration of the new functionalities from the Alkimia library.
- Work([Report](#) 🌐) was on online quote support, equip that with debug functions, extended by an online update of the quote sources, by modifying [libalkimia](#)(backend for KDE finance apps), kmymoney source code (in C++ & Qt) and wrote **unit tests**(using QtTest) with documentation.

Season of KDE student

Remote

KDE Community

Jan 2021 - March 2021

- [Report](#) 🌐 Refactored the python scripts to generate data from docbooks that hugo can populate.

Education

BIT Sindri

2018 – 2022

Bachelor of Technology in Information Technology

CGPA: 8.02/10.00

Technologies

Languages: C, C++, Python, Rust, Java

Technologies: git, bash, Linux, CMake, DevOps (K8s, docker), GTK, Qt

Projects

Autonomous Trading Agent Framework (using TinyLLaMa model)

2025 — [Repo Link](#) 🌐

- An AI agent that: interprets natural language trading strategies (via SLM), converts them into formal strategy code (Rust DSL), integrates to the backtesting engine, and reports results automatically.

C++ Trading Backtester Engine

2025 — [Repo Link](#) 🌐

- A simple event-driven backtesting engine written in C++ for simulating trading strategies using level 1 market data and trade prints.

Simple Java based Physics Engine

2022 — [Repo Link](#) 🌐

- A Java based Physics Engine with functions of Position, Acceleration, Velocity, Vector Mathematics & Collision Dynamics

Smartphone Cluster Computing (Sustainable Technology)

2022 — [Thesis Report](#) 🌐

- A feasibility study on heterogeneous smartphone cluster computing for the major project of the final

semester.

FUSE based file system for Linux

2022 — [Repo Link](#)

- A simple file system in user space that logs user activity.

Port [KDE Timeline](#) Site to Hugo

2020 — [Commit Link](#)

- Mentor - Carl Schwan (KDE eV), Co-Contributor - Phu Nguyen
- [timeline.kde.org](#) is the main site for the history of FOSS and KDE. It used Javascript translations earlier.
- My work was to port all the data to YAML, populate it using a modular approach and perform the i18n translations using hugo translations. We used po4a for the translations

Port [KDE Games](#) Site to Jekyll

2020 — [MR Link](#)

- Created the layouts and ported static pages to markdown.

Miscellaneous

- *Fall 2024-25* **Attendee** [DeFi Talents](#) by [Frankfurt School](#)
- *Dec 2024* **Speaker** [GNOME.Asia Summit](#) Push notification architecture for GNOME Mobile / Phosh
- *Dec 2023* **Speaker** [GNOME.Asia Summit](#) [Contributing to Phosh](#)
- *Sept 2023* **Speaker** [24th Debian Conference](#): [Sustainable Computing](#)
- *March 2022* **Qualified** Graduate Aptitude Test in Engineering (GATE) CS & IT
- *Nov 2020* **Speaker** [Seattle GNU/Linux Conference](#) [Introducing FOSS Culture at Universities](#)

Extra-curricular

- *2020-21* **Vice President** [Hackathon and Coding Club](#) BIT Sindri
- *2020-21* **Operations** [Institution of Electronics and Telecommunication Engineers \(IETE\)](#) forum, BIT Sindri